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Title: 2nd Revised baseline text for X.5Gsec-srocv: Security Requirements for the Operation of 5G Core Network to Support Vertical Services

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Abstract: This TD will be the new baseline text of X.5Gsec-srocv.

This TD is based on C237, C300 and part of C306. The main revisions are as follows:

- Summary is updated.
- Some terms and definitions are added and updated in clause 3.
- Some abbreviations are added in clause 4.
- Chapter 7 is updated.
- The figure of overview of security requirements in clause 8.1 is updated.
- Clause 8.2 is updated.
- Clause 8.3 is restructured and redrafted.
- Clause 8.4 is updated.

Draft Recommendation ITU-T X.5Gsec-srocv5

Security Requirements for the Operation of 5G Core Network to Support Vertical Services

Summary

With the development of 5G system, vertical industries have become key scenarios.

In vertical services for smart factories and smart cities that use an IMT-2020 network, many user's devices use vertical services with massive machine-type communications (mMTC) and ultra-reliable low-latency communications (URLLC).

To support vertical use cases which require ultra reliability and low latency, some of 5G core network's network functions might be deployed at local sites that are closer to vertical users, while other 5G core network's network functions still be deployed at central site. Compared to 5G core network deployed as a whole, 5G core network operating in such deployment might face ~~special~~ specific security threats.

In order to ensure the security of the operation of 5G core network to support vertical services, the specific security threats need to be analysed and relevant security requirements need to be specified.

This ~~recommendation~~ Recommendation aims to analyse data security threats, network security threats and physical security threats specific to the operation of 5G core network deployed at both central site and local sites to support vertical services, and specify the relevant security requirements.

Keywords

5G core network; vertical service; security requirement

Draft Recommendation ITU-T X.5Gsec-srocv5

Security Requirements for the Operation of 5G Core Network to Support Vertical Services

1. Scope

This draft Recommendation focuses on security requirements for the operation of 5G Core Network deployed at central site and local sites to support vertical services. It analyses the data security threats due to operation and subscription data being transferred between central site and local sites, network security threats due to local sites connecting with central site, and physical security threats due to local sites being closer to vertical users. It also, and specifies the relevant security requirements to mitigate the security threats.

2. References

[ITU-T X.1813] Security requirements for the operation of vertical services supporting ultra-reliable and low latency communication (URLLC) in the IMT-2020 private networks

[ITU-T X.5Gsec-netec] Security Capabilities of Network Layer for 5G Edge Computing

3. Definitions

3.1 Terms defined elsewhere

3.1.1 mutual authentication [b-X.1254]: Authentication of identities of entities which provides both entities with assurance of each other's identity.

3.1.2 threat [b-ISO/IEC 27000]: potential cause of an unwanted incident, which can result in harm to a system or organization

3.1.3 vertical service [b-ITU-T X.1813]: From a business perspective, a vertical service is a service focused on a specific industry or group of customers with specialized needs (e.g., automotive services, entertainment services, e-health services, industry 4.0).

3.2 Terms defined in this Recommendation

4. Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

AMF Access and Mobility Management Function

DDoS Distributed Denial of Service

NEF Network Exposure Function

NF Network Function

SMF Session Management Function

UDM Unified Data Management

UPF User Plane Function

URLLC Ultra- reliable and low latency communication

5. Conventions

This Recommendation uses the following conventions:

The keywords “is required to” indicate a requirement which must be strictly followed and from which no deviation is permitted, if conformance to this Recommendation is to be claimed.

The keywords “is recommended” indicate a requirement which is recommended but which is not absolutely required. Thus, this requirement need not be present to claim conformance.

The keywords “is prohibited from” indicate a requirement which must be strictly followed and from which no deviation is permitted, if conformance to this Recommendation is to be claimed.

The keywords “can optionally” indicate an optional requirement which is permissible, without implying any sense of being recommended. This term is not intended to imply that the vendor’s implementation must provide the option, and the feature can be optionally enabled by the network operator/service provider. Rather, it means the vendor may optionally provide the feature and still claim conformance with the specification.

6. Deployment scheme

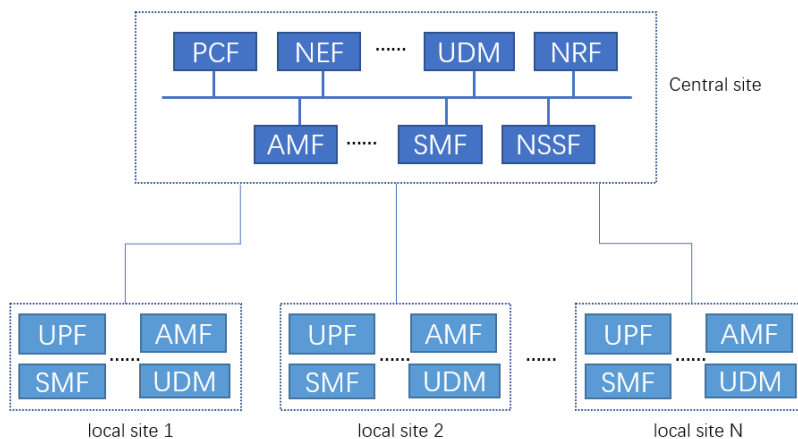


Figure 6-1 Sample deployment of 5G core network to support vertical services

To support vertical services, some of 5G core network’s network functions, like UPF, AMF, SMF, NEF, UDM etc., might be deployed at local sites that are closer to vertical users, while other 5G core network’s network functions still be deployed at central site.

In this deployment, local sites are connected with central sites. And some data will be transfer takes place between central NFs and local NFs. Subscription data like authentication subscription data, access and mobility subscription data, SMF selection subscription data etc. might be transferred between central UDM and local UDM. Operation data like PCC rules, packet flow description, packet detection rules etc. might be transferred between other central NFs and local NFs. Also, subscription data and operation data might be stored in local sites.

7. Security threats

7.1 Data security threats

As data ~~would be~~ transferred between central site and local sites, there are some security threats:

- 1) Data might be eavesdropped, modified, or tampered by attackers during transfer.
- 2) Without proper dispatch control, data related to one vertical user might be transferred to a local site that serves another vertical user.
- 3) A compromised local NF might take access to not only central NF but also local NFs and post fake or malicious data ~~to central NFs~~, or delete or tamper the data in central NFs.
- 4) Data stored in local NFs might be accessed or tampered by attackers.

Since local sites are connected with central site ~~through signal plane network and management plane network~~, ~~that local NFs can access central NFs, and NFs from one local site might access NFs from another local site through central site network.~~ might be some security threats.

Security threats from local NFs to central NFs include:

- 1) Compromised local NFs might scan to find out central NFs' vulnerabilities and launch further attacks.
- 2) Compromised local NFs might launch DDoS attacks to central sites.
- 3) If there is not mutual authentication between local NFs and central NFs, fake local NFs might access central NFs and interfere with central NFs' operation.

Security threats from different local sites' NFs include:

- 2) Compromised NFs at one local site might launch DDoS attacks to other local sites.
- 3) ...

~~If local NFs' credential is not properly protected, attackers could get the credential and fake local NF to communicate with central site.~~

Since local NFs are closer to vertical users, they are more likely to be physically attacked.

- 1) Malicious patches/programs might be installed in local NFs.
- 2) Attackers might control local NFs through local maintenance interfaces.
- 3) Attackers might reset local NFs' passwords by making them enter recovery mode.
- 4) Attackers might also access physically and tamper the local NFs.
- 5) ...

8. Security requirements

~~[This chapter specifies security requirements for the operation of 5G core network deployed at both central site and local sites to support vertical services, which include data security requirements to mitigate the threats due to operation and subscription data being transferred between central site and local sites; network security requirements to mitigate the threats due to local sites connecting with central site; physical security requirements to mitigate the threats due to local sites being closer to vertical users.]~~

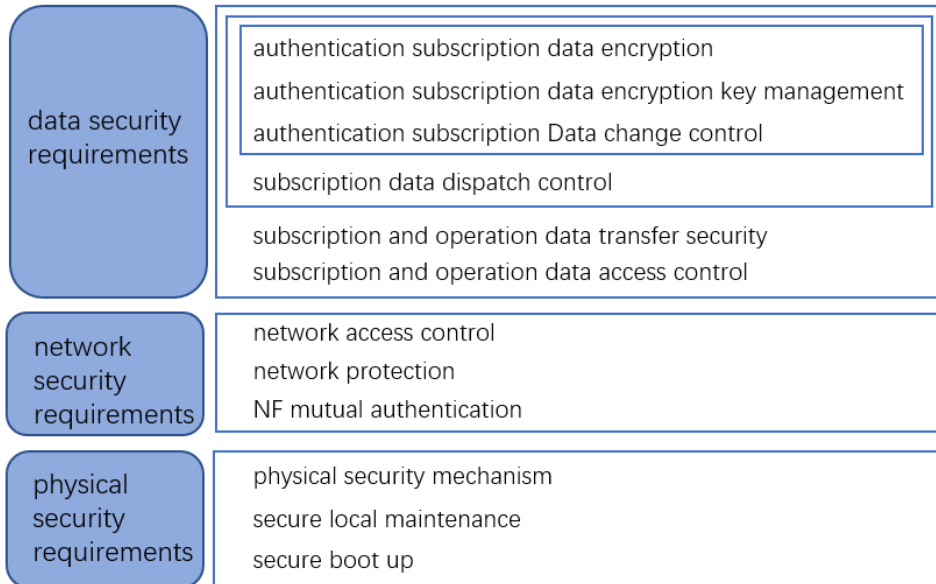


Figure8-1 draft overview of security requirements for the operation of 5G core network deployed at both central site and local sites to support vertical services

Authentication subscription data is ~~the data that is~~ needed for the generation of authentication vectors in the UDM, which may consist of long term key K, sequence number SQN, authentication method used, etc. Since authentication subscription data is essential, it requires the protection which include encryption, integrity, entity and source authentication and their associated encryption key management and change control.

8.2 Data security requirements

The security requirements for authentication subscription data include:

- 1) It is required that Authentication subscription data is encrypted when it is exported out of central UDM.
- 2) It is required that different encryption keys are used to encrypt different local sites' authentication subscription data.
- 3) ~~It is required that~~ And the cryptographic keys for encryption, integrity, entity authentication keys should be securely protected.
- 4) It is required that only authorized NFs can change authentication subscription data stored in local UDM.
- 5) It is required that local NFs cannot change authentication subscription data stored in central UDM.

The security requirements for all the subscription data include:

- 1) It is required that subscription data is only ~~be~~ transferred from central site to local sites. Transferring subscription data from local sites to central site should be prohibited.
- 2) It is required that subscription data of one local site cannot be transferred to other local sites.

The security requirements for both subscription data and operation data include:

- 1) It is required that subscription and operation data are integrity protected during transfer.
- 2) It is required that access control is applied to subscription and operation data stored in local sites.

~~And the encryption keys should be securely protected.~~

8.3.1 Network access control

The security requirements for network access control between local NFs to central NFs include:

- 1) It is required that only specific IP addresses of local sites are permitted to access central site.
- 2) It is required that local NFs are only permitted to access related central NFs. For example, local UPF can access central SMF, but local UPF should be prohibited from accessing central AMF.
- 3) It is recommended that local NFs are only permitted to require central NFs' related services. For example, local AMF can require central UDM's Nudm_UECM_Registration service, but local AMF should be prohibited from requiring central UDM's Nudm_EventExposure_Subscribe service.

The security requirements for network access control between different local sites include:

- 1) It is required that local sites serving different vertical users cannot access each other.

~~8.3.1.1 Access control at signal plane network~~

~~There should be access controls between local sites, so that local sites serving different vertical users cannot access each other.~~

The security requirements for network protection include:

- 1) It is recommended that abnormal packets from local sites to central site can be detected and abandoned.
- 2) It is recommended that there are intrusion detection/prevention mechanisms between local sites and central site.
- 3) ~~It is recommended that there should be~~ occur as a result of a denial-of-service attack or during periods of increased traffic. In particular, partial or complete impairment of system availability shall be avoided. This can be done through proper rate limits between local site and central site, so that compromised local NFs cannot launch DDoS attacks toward central NFs.
- 4) It is recommended that there should be a logging mechanism in such a way that security events shall be logged together with a unique system reference (e.g. host name, IP or MAC address) along with exact time of incident.

~~There should be proper rate limits between local site and central site, so that compromised local NFs cannot launch DDoS attacks toward central NFs.~~

~~8.3.3 Abnormal packet filter~~

~~8.3.4~~ 3 NF mutual authentication

The security requirements for NF mutual authentication include:

- 1) It is required that mutual authentication should be performed between local NFs and central NFs.
- 2) It is required that NFs' credentials should be securely protected.
- 3) It is required that Strong password policy along with protection against brute force and dictionary attacks should be implemented.
- 4) It is recommended that the authorisations for accounts and applications shall be reduced to the minimum required for the tasks they have to perform. Authorisations to a system shall be restricted to a level and access be controlled to a specific user.

~~Mutual authentication should be performed between local NFs and central NFs.~~

~~NFs' credential should be securely protected.~~

Unused software components or parts of software which are not needed for operation or functionality of the network product should not be installed or be deleted after installation.

8.4.1 physical security mechanism

There should be physical security mechanism to protect local NFs from being stolen, turned off etc.

The physical security which can be implemented are as follows:

- i. Access of local NF should be restricted physically and be secured through Electro-Mechanical locks with strong PIN.
- ii. Physical location of NF should not be easily accessible.
- iii. Physical accessible interfaces which are not under use shall be disabled permanently.

...

8.4.2 Secure local maintenance

It should be prohibited to recover local NFs' system password by entering recovery mode.

Stringent Password policy rules must be implemented.

Access control mechanism must be implemented.---

...

8.4.3 Secure boot up

Local NFs should not boot up from external storage.

...
